

Invariant sums of random matrices and the onset of level repulsion

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Abstract. Using a simple rotate-and-sum procedure, we construct and solve exactly a random matrix model with peculiar features. It is invariant under the standard symmetry groups (orthogonal and unitary) and yet the interaction between eigenvalues is not Vandermondean. The ensemble contains real symmetric or complex hermitian matrices $\mathbf{S}$ of the form $\mathbf{S} = \sum_{i=1}^M \langle \mathbf{O}_i \mathbf{D}_i \mathbf{O}_i^T \rangle$ or $\mathbf{S} = \sum_{i=1}^M \langle \mathbf{U}_i \mathbf{D}_i \mathbf{U}_i^\dagger \rangle$ respectively. The diagonal matrices $\mathbf{D}_i = \text{diag}\{\lambda_1^{(i)}, \dots, \lambda_N^{(i)}\}$ are constructed from real eigenvalues drawn *independently* from distributions $p^{(i)}(x)$, while the matrices $\mathbf{O}_i$ and $\mathbf{U}_i$ are all orthogonal or unitary and the average $\langle \cdot \rangle$ is performed over the respective group. While the original matrices $\mathbf{D}_i$ do not exhibit level repulsion, the resulting sum $\mathbf{S}$ develops it upon averaging over multiple ($M \geq 2$) uncorrelated rotations. We focus on the cases where $p^{(i)}(x)$ is A.) a semicircle law, or B.) a Gaussian law for all $i = 1, \dots, M$. For the choice A, in the limit $N \rightarrow \infty$ this ensemble appears spectrally indistinguishable from the standard GOE or GUE, having same spectral density, two-point correlation function, and nearest-neighbor spacing distribution $p(s)$ after unfolding. However, working out the case $N = 2$ in detail, we uncover a *universal* (independent of the $p^{(i)}(x)$) but different from Wigner-Dyson behavior as $s \rightarrow 0^+$. The generic interaction between eigenvalues of $\mathbf{S}$ is indeed *not* precisely Vandermondean, despite the rotationally invariant nature of the ensemble, and classical RMT universality is restored only asymptotically. This construction allows to probe analytically the microscopic statistics of free addition of random matrices. The analytical results are corroborated by numerical simulations.

1. Foreword

Consider the following three plots in Fig. 1. The numerical points represent (clockwise) the average spectral density, the nearest-neighbor spacing distribution and the two-point correlation function of (unfolded) eigenvalues for a numerically generated ensemble of large random matrices with unitary and orthogonal invariance.

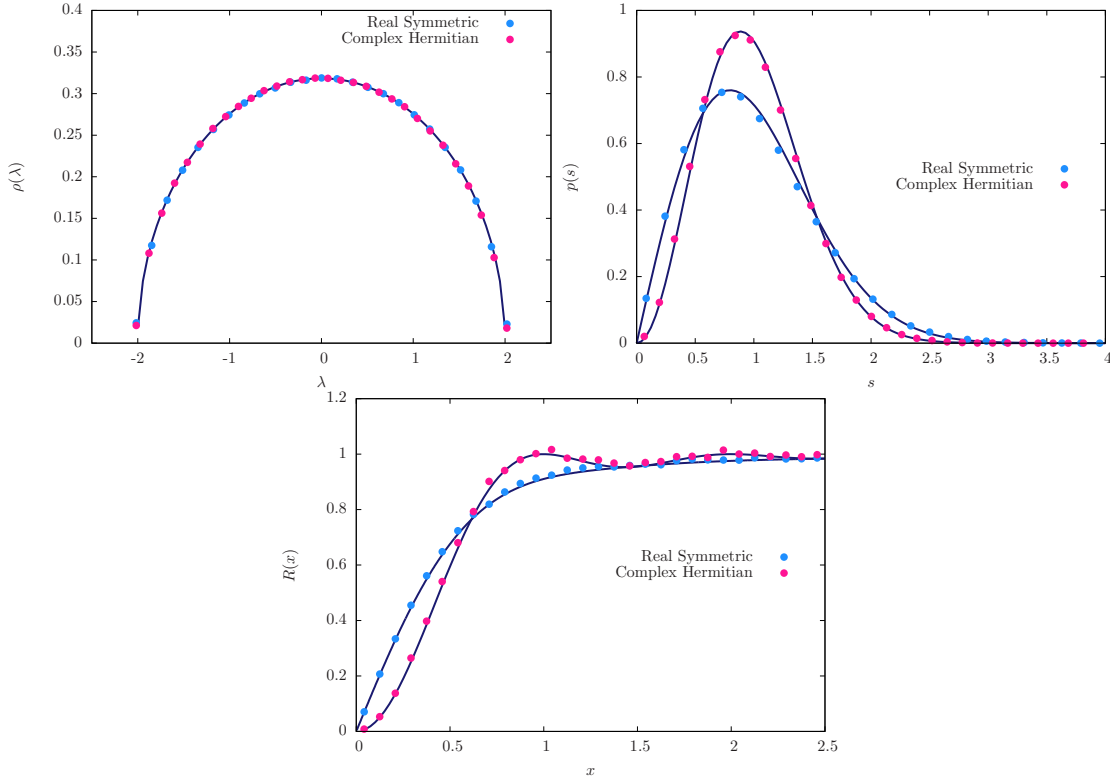


Figure 1. Numerical points: Spectral density (top left), level spacing distribution (top right) and two-point correlation function (bottom) for a numerically generated model of real symmetric (complex hermitian) 500×500 random matrices. In all plots, solid lines refer instead to the large N results for the corresponding quantities of the GOE and GUE.

We challenge the reader to guess what this ensemble is, based on the figures at hand.

2. Introduction

Since its inception in nuclear physics more than sixty years ago, and much earlier in statistics, Random Matrix Theory (RMT) has become an invaluable tool in many fields of physics and mathematics, with countless applications (see [1, 2, 3, 4]). Traditionally, among the ensembles of random matrices with real spectra for which an analytical treatment (to some extent) is feasible we may single out

- Matrices with independent identically distributed (i.i.d.) entries (Wigner matrices) [5], such as adjacency matrices of Erdős-Rényi graphs [6].
- Matrices with rotational invariance, where the joint probability density (jpd) of the entries $\mathcal{P}[\mathbf{H}] \equiv \mathcal{P}(H_{11}, \dots, H_{NN})$ remains unchanged if one performs a similarity transformation $\mathbf{H} \rightarrow \mathbf{O}\mathbf{H}\mathbf{O}^T$ (for real symmetric matrices $\mathbf{H}$, $\mathbf{O}$ being an arbitrary orthogonal matrix), $\mathbf{H} \rightarrow \mathbf{U}\mathbf{H}\mathbf{U}^\dagger$ (for complex hermitian matrices $\mathbf{H}$, $\mathbf{U}$ being an arbitrary unitary matrix) or $\mathbf{H} \rightarrow \mathbf{\Sigma}\mathbf{H}\mathbf{\Sigma}^*$ (for quaternion self-dual matrices $\mathbf{H}$, $\mathbf{\Sigma}$

being an arbitrary symplectic matrix). Here, $(\cdot)^T$, $(\cdot)^\dagger$ and $(\cdot)^*$ stand for transpose, hermitian conjugate and symplectic conjugate of the matrix respectively.

For such matrices, the joint probability density of eigenvalues $\mathcal{P}(\lambda_1, \dots, \lambda_N)$ can be generically written as

$$\mathcal{P}(\lambda_1, \dots, \lambda_N) \propto e^{-\beta N \sum_{i=1}^N V(\lambda_i)} |\Delta(\boldsymbol{\lambda})|^\beta, \quad (1)$$

where $\Delta(\boldsymbol{\lambda}) = \prod_{j < k} (\lambda_j - \lambda_k)$ is the Vandermonde determinant, $V(x)$ is a potential suitably growing at infinity, and the Dyson index $\beta = 1, 2, 4$ for real symmetric, complex hermitian and quaternion self-dual matrices respectively[‡]. In this case, the eigenvectors are uniformly distributed (with Haar measure) in the corresponding symmetry group (orthogonal, unitary or symplectic respectively).

The only ensemble with independent entries and rotational invariance is the Gaussian ensemble ($V(x) = x^2/2$), where the entries in the upper triangle are independently sampled from a Gaussian distribution (in the real, complex or quaternion domain). These ensembles are then called Gaussian Orthogonal Ensemble (GOE), Gaussian Unitary Ensemble (GUE) and Gaussian Symplectic Ensemble (GSE) respectively.

The presence of the Vandermonde determinant $\prod_{j < k} (\lambda_j - \lambda_k)$ in (1) implies that the eigenvalues of invariant matrix models are strongly correlated variables, whose statistics is very different from the i.i.d. case. In particular, the distribution of spacings $p(s)$ between adjacent eigenvalues generally develops a behavior like $p(s) \sim s^\beta$ for $s \rightarrow 0$, which is known as *level repulsion*. Once the overall density of eigenvalues is discounted from numerically diagonalized ensembles through a procedure called *unfolding* (see [Appendix A](#)), this repulsion is largely independent of the particular choice of the confining potential, one of the very many manifestations of *universality* in RMT. The tendency of eigenvalues to repel each other is markedly different from the case of uncorrelated random variables on an interval, which tend to cluster and develop a spacing distribution of the Poisson form $p(s) \sim e^{-s}$. In the field of quantum chaos, such different statistics for the spacings between energy levels help discriminating between quantum systems whose classical counterpart is chaotic or integrable [7]. It is therefore one of the central observables in the so-called *microscopic* analysis of spectra.

Indeed, in RMT one usually makes a distinction between *macroscopic* and *microscopic* spectral properties. While the former take into account the global distribution of the eigenvalues (on a scale much larger than the mean level spacing) and are generally non-universal (highly dependent on the invariant ensemble under consideration), the latter probe the spectral statistics on the scale of the mean inter-level distance and have a high degree of universality across ensembles for large matrix size. Typical examples of the two classes of observables are the average spectral density (macroscopic, non-universal)

$$\rho_\infty(\lambda) = \lim_{N \rightarrow \infty} \rho_N(\lambda), \quad \text{where} \quad \rho_N(\lambda) = \left\langle \frac{1}{N} \sum_{i=1}^N \delta(\lambda - \lambda_i) \right\rangle, \quad (2)$$

[‡] We will not consider the quaternion case ($\beta = 4$) here.

(here $\langle \cdot \rangle$ stands for averaging with respect to the jpd (1)) and the level spacing distribution and the two-point correlation function (microscopic, universal: see [Appendix A](#)). However, pairs of *different* ensembles that share *all* the following features

- they are both rotationally invariant,
- they have the same (macroscopic) spectral density for $N \rightarrow \infty$,
- they have the same level spacing distribution and two-point correlation function (microscopic) for $N \rightarrow \infty$,

have not been identified so far, to the best of our knowledge. It is one of the goals of this paper to produce an explicit example of such “spectrally indistinguishable” ensembles (even if such indistinguishability does not extend to higher-order correlation functions). Besides constituting a rather exotic mathematical curiosity, albeit yielding an exactly solvable model, the construction we propose gives also analytical access to the microscopic statistics of the *free* addition of random matrices - a rare and very welcome luxury.

The framework of *free probability* is a powerful extension of the concept of independence for random variables to non-commutative objects. Pioneered by Voiculescu [8], it has found natural applications in the field of random matrices [9] where it allows to compute the average density of eigenvalues (in the large matrix size limit $N \rightarrow \infty$) for sums or product of random matrices enjoying a property called *freeness*. While its precise mathematical definition is complicated [10], it can be roughly identified with the simultaneous occurrence of the following features for the matrices being summed:

- Independence of different matrices.
- No correlations between the eigenspaces of different matrices.
- Asymptotic limit of large matrix size, $N \rightarrow \infty$.

If these conditions are satisfied, a general procedure exists to compute the average spectral density of sums (or products) of free random matrices of infinite size, starting from the spectral densities of the individual summands (or factors) (see [11] for a recent review). However, more detailed spectral information (encoded for instance in the jpd of eigenvalues, or the microscopic statistics) is generally unavailable, as well as finite N results of any sort - with the exception of very recent developments for the products of Ginibre and Wishart matrices [12, 13]. In this respect, our model offers a unified (and otherwise unavailable) view on the features of a free sum, macroscopic universality in the $N \rightarrow \infty$ limit (checked numerically) as well as microscopic properties (addressed analytically for small N and numerically for large N). We start by first defining the model.

3. The model

We consider a rotationally invariant ensemble of real symmetric or complex hermitian $N \times N$ matrices $\mathbf{S}$ of the form

$$\mathbf{S} = \sum_{i=1}^M \langle \mathbf{O}_i \mathbf{D}_i \mathbf{O}_i^T \rangle, \quad \mathbf{S} = \sum_{i=1}^M \langle \mathbf{U}_i \mathbf{D}_i \mathbf{U}_i^\dagger \rangle, \quad (3)$$

respectively. The diagonal matrices $\mathbf{D}_i = \text{diag}\{\lambda_1^{(i)}, \dots, \lambda_N^{(i)}\}$ are built out of independent and identically distributed real eigenvalues drawn from distributions $p^{(i)}(x)$, while the matrices $\mathbf{O}_i$ and $\mathbf{U}_i$ are (uncorrelated) orthogonal or unitary matrices (respectively), and the average $\langle \cdot \rangle$ is performed over the respective group with a Haar measure.

Rotational invariance is proven (for simplicity in the complex hermitian case) in Section 5, where we also provide the jpd of eigenvalues $\mathcal{P}(\boldsymbol{\nu}) \equiv \mathcal{P}(\nu_1, \dots, \nu_N)$ for this model in the complex hermitian (unitary) case

$$\mathcal{P}(\boldsymbol{\nu}) \propto \Delta(\boldsymbol{\nu}) \left\langle \int \prod_{j=1}^N dt_j \frac{\det(e^{-i\nu_j t_k})_{j,k=1 \rightarrow N}}{[\Delta(\mathbf{t})]^{M-1}} \prod_{i=1}^M \frac{\det(e^{i\lambda_j^{(i)} t_k})_{j,k=1 \rightarrow N}}{\Delta(\boldsymbol{\lambda}^{(i)})} \right\rangle_{\{p^{(i)}(\lambda)\}}, \quad (4)$$

which has a quite unusual “multi-orthogonal” form (the average here $\langle \cdot \rangle_{\{p^{(i)}(\lambda)\}}$ stands for $\int \prod_{j=1}^N \prod_{i=1}^M d\lambda_j^{(i)} p^{(i)}(\lambda_j^{(i)})$). If in addition the distribution of eigenvalues is the same for all summands, $p^{(i)}(x) \equiv p(x)$, $\forall i$, Eq. (4) simplifies to

$$\mathcal{P}(\boldsymbol{\nu}) \propto \Delta(\boldsymbol{\nu}) \int \prod_{j=1}^N dt_j \frac{\det(e^{-i\nu_j t_k})_{j,k=1 \rightarrow N}}{[\Delta(\mathbf{t})]^{M-1}} \left[\int \prod_{j=1}^N d\lambda_j p(\lambda_j) \frac{\det(e^{i\lambda_j t_k})_{j,k=1 \rightarrow N}}{\Delta(\boldsymbol{\lambda})} \right]^M. \quad (5)$$

Note that the case $M = 1$ is trivial, as it corresponds to a simple “averaged” similarity transformation (so the spectra of $\mathbf{S}$ and $\mathbf{D}_1$ are identical, see also Section 5). For $M \geq 2$ and $N \gg 1$, assuming the same distribution of $\boldsymbol{\lambda}$ -eigenvalues for all M -matrices (notably we focus on the semicircle law and the Gaussian law as explicit examples), we are precisely in the typical setting where free probability can be applied (large independent matrices, and no correlation between eigenspaces of the summands). The interesting new point is that we are now able to investigate *both* the macroscopic *and* the microscopic spectral statistics of the model. Here we give a summary of the main features and the corresponding Sections in the paper.

- **Macroscopic:** clearly, the macroscopic spectral properties are univocally determined by the distribution $p^{(i)}(x)$, provided that we choose the same one for all i 's. For example, in the Foreword we have constructed our ensemble as $\mathbf{D}_1 + \mathbf{O} \mathbf{D}_2 \mathbf{O}^T$ ($\mathbf{D}_1 + \mathbf{U} \mathbf{D}_2 \mathbf{U}^\dagger$), where $\mathbf{D}_{1,2}$ are diagonal matrices filled with independent and identically distributed elements drawn from a semicircle distribution

$$p^{(i)}(x) = \frac{1}{2\pi} \sqrt{4 - x^2} \quad \forall i, \quad (6)$$

whereas $\mathbf{O}$ ($\mathbf{U}$) is a random Haar orthogonal (unitary) matrix (we used $N = 500$ and $M = 2$ there)[§]. The average spectral density of the model is again, of course, the semicircle distribution (see Fig. 1). Note, however, that (at odds with the Gaussian case) the edges of the semicircle are *hard* in our model: even for finite N , it is by construction impossible to sample eigenvalues exceeding the edges. It would be also interesting to study the spectral density of the matrix sum $\mathbf{S}$ when different distributions $p^{(i)}(x)$ for different i 's are chosen. This, however, goes beyond the scope of the present paper, and is left for future investigation.

- **Microscopic:** in Section 4, we analyze the nearest-neighbor spacing distribution for real symmetric and complex Hermitian matrices starting from the definition in (3), for an arbitrary choice of the distributions $p^{(i)}(x)$. We show that the spacing distribution $p(s)$ between the two eigenvalues (ν_1, ν_2) , $s = \nu_2 - \nu_1$, for $N = M = 2$ is (as expected) entirely different from the Gaussian case, where the distribution is given by the so-called *Wigner's surmise* $p_{WS}^{(\beta)}(s)$ (see Appendix A for details about its universal properties). However, the $p(s)$ of our model still retains a certain degree of universality (*i.e.* independence on the specific form of the distributions $p^{(i)}(x)$). Indeed, the behavior of $p(s)$ for small spacing s is universally given by $p(s) \approx -s \ln s$ in the orthogonal case (see equation (24)), and $p(s) \approx s$ in the unitary case (see equation (41)), with prefactors that can be computed exactly in full generality. In Section 5, we focus on the unitary case, proving rotational invariance of the ensemble and computing the jpd (4). Next, in Section 5.1 we use this result to compute $p(s)$ exactly for $N = M = 2$ (clearly for the unitary case, with a Gaussian distribution $p^{(i)}(x) = e^{-x^2/2}/\sqrt{2\pi} \forall i$), this time starting from the jpd (4), and we recover the general behavior at $s \rightarrow 0^+$. Still for the unitary case with a Gaussian law, in Section 5.2 we analyze the case $M > 2$ for $N = 2$: we find yet another behavior at small spacing (see Table 1). For large enough N , no matter how many matrices M we sum, we recover the “usual” universal behaviors (in the Wigner's surmise form $p_{WS}^{(\beta)}(s)$ for $\beta = 1, 2$) for invariant ensembles after unfolding (see Fig. 1 and Appendix A). In Sec. 6, we study the spacing distribution for $N = 2$ and $M \gg 1$, using an elegant mapping to the problem of summing randomly oriented vectors in two and three dimensions. In Table 1, we summarize the interplay between N and M and our findings in each situation.

Analyzing different distributions $p^{(i)}(x)$ (cases not shown), we are able to formulate the conjecture that the nearest-neighbor spacing distribution and the two-point correlations functions *computed on unfolded eigenvalues* have the universal GOE or GUE form for our model in the limit $N \rightarrow \infty$, irrespective of the marginal distributions $p^{(i)}(x)$ and the number of summands $M \geq 2$. Proving this conjecture starting from the jpd (4) does not seem a trivial task, though.

[§] In order to generate random Haar orthogonal and unitary matrices we employed the algorithm described in [14].

Summary		
	$N = 2$	$N \gg 1$
$M = 2$	$p(s) \approx -s \ln s$ (<u>Orthogonal</u>) (24) $p(s) \approx s$ (<u>Unitary</u>) (41) (see Section 4)	Standard Wigner-Dyson universality (See Fig. 1 for semicircle law)
$M > 2$	<u>Unitary, Gaussian law</u> $M = 3$: $p(s) \approx -s^2 \ln s$ (75) $M \geq 4$: $p(s) \approx s^2$ (73) <u>Unitary and Orthogonal cases</u> $M \gg 1$: scaling forms for $p(s)$ (97),(107)	Same behavior as in Fig. 1 (not shown)

Table 1. Schematic summary of the interplay between N (size of the matrices) and M (number of summands). For $N = M = 2$, the behavior of the spacing distribution for $s \rightarrow 0^+$ can be worked out in full generality (irrespective of the distributions $p^{(i)}(x)$ (see Section 4)). For $N \gg 1$, we investigated numerically the semicircle law and found that classical RMT universality is restored (see Fig. 1), irrespective of the number of summands ($M \geq 2$). We conjecture that this RMT universality for $N \gg 1$ would be recovered irrespective of the distributions $p^{(i)}(x)$. Finally, for $N = 2$ in the unitary case and for a Gaussian distribution of eigenvalues, we find a different behavior of the spacing distribution for $M = 3$ and $M \geq 4$, as shown in the Table. Finally, for $N = 2$ and $M \gg 1$, we can invoke a multidimensional version of the central limit theorem to show that the spacing distribution attains scaling forms in the orthogonal and unitary cases (see Section 6), irrespective of the distribution of diagonal matrices.

In the Appendix, we provide a summary of the unfolding procedure for nearest-neighbor spacings and textbooks formulae for two-point correlation functions that are used for comparison with numerical simulations.

4. Level Spacings $p(s)$ for $N = M = 2$ and universal behavior for $s \rightarrow 0^+$

In this section, we compute analytically the probability density $p(s)$ of the level spacing $s = \nu_2 - \nu_1$ for the case $N = M = 2$, for real symmetric and complex hermitian matrices $\mathbf{S}$. We offer here a derivation uniquely based on the matrix representation (3). This allows to i.) treat the orthogonal and unitary cases on the same footing, and ii.) to cross-check the result for the unitary case against a derivation based on the jpd (4) (see Section 5). We assume without loss of generality the following form for the matrix sum

$$\mathbf{S} = \mathbf{D}_1 + \langle \mathbf{U} \mathbf{D}_2 \mathbf{U}^\dagger \rangle, \quad \mathbf{S} = \mathbf{D}_1 + \langle \mathbf{O} \mathbf{D}_2 \mathbf{O}^T \rangle, \quad (7)$$

where $\mathbf{U}$ is a unitary 2×2 matrix, $\mathbf{O}$ is a orthogonal 2×2 matrix and $\langle \cdot \rangle$ stands for integration over the respective group. For simplicity, we rename the eigenvalues as (λ_1, λ_2) for the diagonal matrix $\mathbf{D}_1$ and (μ_1, μ_2) for $\mathbf{D}_2$. They are independently drawn from probability densities $p^{(1)}(\lambda), p^{(2)}(\mu)$ respectively. The matrix $\mathbf{U}$ can be parametrized as

$$\mathbf{U} = \begin{pmatrix} e^{-i\phi_1} \cos(t) & -e^{-i\theta-i\phi_1} \sin(t) \\ e^{i\theta+i\phi_2} \sin(t) & e^{i\phi_2} \cos(t) \end{pmatrix} \quad (8)$$

for the unitary case, and

$$\mathbf{O} = \begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix} \quad (9)$$

for the orthogonal case, where $0 \leq \phi_1, \phi_2, \theta \leq 2\pi$ and $0 \leq t \leq \pi/2$. It turns out that the spacing s between the largest and the smallest eigenvalue of $\mathbf{S}$ has an expression that depends only on the angle t and is the same for the unitary and orthogonal cases, $s = \sqrt{q}$, where

$$q = (\lambda_2 - \lambda_1)^2 + (\mu_2 - \mu_1)^2 + 2(\lambda_2 - \lambda_1)(\mu_2 - \mu_1) \cos(2t) . \quad (10)$$

The probability density of q is therefore given by

$$\mathcal{P}(q) = \frac{2}{\pi} \int_{\sigma_1} d\lambda_1 d\lambda_2 p^{(1)}(\lambda_1) p^{(1)}(\lambda_2) \int_{\sigma_2} d\mu_1 d\mu_2 p^{(2)}(\mu_1) p^{(2)}(\mu_2) \int_0^{\pi/2} dt \times \\ \times \delta(q - [(\lambda_2 - \lambda_1)^2 + (\mu_2 - \mu_1)^2 + 2(\lambda_2 - \lambda_1)(\mu_2 - \mu_1) \cos(2t)]) , \quad (11)$$

for the orthogonal case, and

$$\mathcal{P}(q) = 2 \int_{\sigma_1} d\lambda_1 d\lambda_2 p^{(1)}(\lambda_1) p^{(1)}(\lambda_2) \int_{\sigma_2} d\mu_1 d\mu_2 p^{(2)}(\mu_1) p^{(2)}(\mu_2) \int_0^{\pi/2} dt \cos(t) \sin(t) \times \\ \times \delta(q - [(\lambda_2 - \lambda_1)^2 + (\mu_2 - \mu_1)^2 + 2(\lambda_2 - \lambda_1)(\mu_2 - \mu_1) \cos(2t)]) , \quad (12)$$

for the unitary case, where the first four integrals run over the supports σ_1 and σ_2 of each probability density (in general distinct). Note that for $M \geq 3$, instead, the expression for q is different in the orthogonal and unitary cases, but it turns out to have a simple interpretation in terms of sums of M random two-dimensional (orthogonal case) or three-dimensional (unitary case) vectors (see Sec. 6).

Then the spacing distribution in both cases is given by

$$p(s) = 2s \mathcal{P}(s^2) . \quad (13)$$

In order to get a manageable expression for $\mathcal{P}(q)$, we first introduce the two identities

$$1 = \int_{-\infty}^{\infty} ds_1 \delta(s_1 - (\lambda_2 - \lambda_1)) \quad (14)$$

$$1 = \int_{-\infty}^{\infty} ds_2 \delta(s_2 - (\mu_2 - \mu_1)) , \quad (15)$$

as well as the following auxiliary functions

$$\omega^{(1)}(s) = \int_{\sigma_1} d\lambda_1 \int_{\sigma_1} d\lambda_2 \delta(s - (\lambda_2 - \lambda_1)) p^{(1)}(\lambda_1) p^{(1)}(\lambda_2) \quad (16)$$

$$\omega^{(2)}(s) = \int_{\sigma_2} d\mu_1 \int_{\sigma_2} d\mu_2 \delta(s - (\mu_2 - \mu_1)) p^{(2)}(\mu_1) p^{(2)}(\mu_2) . \quad (17)$$

Now we consider the two cases (orthogonal and unitary) separately.

4.1. Orthogonal case

We can now rewrite (11) as

$$\mathcal{P}(q) = \frac{2}{\pi} \int_{-\infty}^{\infty} ds_1 ds_2 \int_0^{\pi/2} dt \delta(q - [s_1^2 + s_2^2 + 2s_1 s_2 \cos(2t)]) \omega^{(1)}(s_1) \omega^{(2)}(s_2) . \quad (18)$$

Everything is therefore expressed only in terms of the auxiliary functions $\omega^{(1)}(s)$ and $\omega^{(2)}(s)$ which represent a sort of spacing distributions (including sign!) of the individual diagonal matrices $\mathbf{D}_1$ and $\mathbf{D}_2$. To make further progress, it is convenient to introduce the change of variables $s_1 = (x - y)/2$ and $s_2 = (x + y)/2$, yielding

$$\begin{aligned} \mathcal{P}(q) &= \frac{1}{\pi} \int_{-\infty}^{\infty} dx dy \omega^{(1)}\left(\frac{x-y}{2}\right) \omega^{(2)}\left(\frac{x+y}{2}\right) \times \\ &\times \int_0^{\pi/2} dt \delta\left(q - \frac{1}{2}(x^2 + y^2) - \frac{1}{2}(x^2 - y^2) \cos(2t)\right) . \end{aligned} \quad (19)$$

Setting $\cos(2t) = \xi$, one obtains

$$\begin{aligned} \mathcal{P}(q) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} dx dy \omega^{(1)}\left(\frac{x-y}{2}\right) \omega^{(2)}\left(\frac{x+y}{2}\right) \times \\ &\times \int_{-1}^1 \frac{d\xi}{\sqrt{1-\xi^2}} \frac{\delta\left(\xi - \frac{q - (1/2)(x^2+y^2)}{(1/2)(x^2-y^2)}\right) \mathbb{1}\left(-1 \leq \frac{q - (1/2)(x^2+y^2)}{(1/2)(x^2-y^2)} \leq 1\right)}{(1/2)|x^2 - y^2|} , \end{aligned} \quad (20)$$

where $\mathbb{1}(x)$ is the indicator function, equal to 1 if x is logically true and 0 otherwise. Resolving the constraints, we can express $\mathcal{P}(q)$ as the sum of four contributions

$$\mathcal{P}(q) = \frac{1}{2\pi} \sum_{j=1}^4 I_j(q), \quad I_j(q) = \int_{X_j} dx \int_{Y_j} dy f_q(x, y) , \quad (21)$$

where

$$f_q(x, y) = \frac{\omega^{(1)}\left(\frac{x-y}{2}\right) \omega^{(2)}\left(\frac{x+y}{2}\right)}{\sqrt{(x^2 - q)(q - y^2)}} \quad (22)$$

and $X_1 = Y_2 = (-\infty, -\sqrt{q})$, $X_2 = X_3 = Y_1 = Y_4 = (-\sqrt{q}, \sqrt{q})$ and $X_4 = Y_3 = (\sqrt{q}, \infty)$. It is easy to see that if $\omega^{(1)}(x) = \omega^{(1)}(-x)$, then $I_1(q) = I_2(q)$ and $I_3(q) = I_4(q)$, and if $\omega^{(1)}(x) = \omega^{(2)}(x)$, then $I_2(q) = I_3(q)$.

Therefore, if $\omega^{(1)}(x) = \omega^{(2)}(x)$ is an even function, then $\mathcal{P}(q)$ can be simplified as

$$\mathcal{P}(q) = \frac{1}{2\pi} \times 4 \int_{X_j} dx \int_{Y_j} dy f_q(x, y), \quad \text{any } j . \quad (23)$$

We shall mainly restrict to this case henceforth. Furthermore if $0 < \omega^{(1)}(0) < \infty$, we prove that for $M = N = 2$ the level repulsion at zero is universally given by the following non-Wigner behavior

$$p(s) \sim -4[\omega^{(1)}(0)]^2 s \ln s, \quad s \rightarrow 0^+, \quad (24)$$

a direct consequence of the fact that the repulsion between eigenvalues of $\mathbf{S}$ is not precisely Vandermonde, as in the standard invariant ensembles. A similar, weaker repulsion of eigenvalues was detected in another random matrix models for pseudo-Hermitian matrices [15]. In the asymptotic limits $M \geq 2, N \rightarrow \infty$, however, the standard Wigner-surmise behavior is recovered after unfolding (see Fig. 1, top right, for the case $M = 2$).

In order to prove (24), we start from (23) (combined with (13)) in the form

$$p(s) = \frac{4}{\pi} s \int_s^\infty dx \int_{-s}^s dy \frac{\omega^{(1)}\left(\frac{x-y}{2}\right) \omega^{(1)}\left(\frac{x+y}{2}\right)}{\sqrt{(x^2 - s^2)(s^2 - y^2)}}. \quad (25)$$

For small s , the integral in y can be estimated as

$$\int_{-s}^s dy \frac{\omega^{(1)}\left(\frac{x-y}{2}\right) \omega^{(1)}\left(\frac{x+y}{2}\right)}{\sqrt{s^2 - y^2}} \rightarrow \left[\omega^{(1)}\left(\frac{x}{2}\right)\right]^2 \int_{-s}^s \frac{dy}{\sqrt{s^2 - y^2}} = \pi \left[\omega^{(1)}\left(\frac{x}{2}\right)\right]^2. \quad (26)$$

Therefore

$$p(s) \sim 4s \int_s^\infty dx \frac{\left[\omega^{(1)}\left(\frac{x}{2}\right)\right]^2}{\sqrt{x^2 - s^2}}. \quad (27)$$

Any singular behavior of the integral (44) can only arise in the vicinity of $x \simeq s$. Therefore, for s close to zero we can estimate the integral contribution as

$$\int_s^\infty dx \frac{\left[\omega^{(1)}\left(\frac{x}{2}\right)\right]^2}{\sqrt{x^2 - s^2}} \rightarrow [\omega^{(1)}(0)]^2 \int_s^1 \frac{dx}{\sqrt{x^2 - s^2}} = [\omega^{(1)}(0)]^2 \ln \left(\frac{1 + \sqrt{1 - s^2}}{s} \right). \quad (28)$$

Expanding the logarithm around $s = 0$, and collecting prefactors we precisely arrive at (24). We can verify this general statement by explicitly drawing the eigenvalues of the matrices $\mathbf{D}_1$ and $\mathbf{D}_2$ e.g. from a standard normal distribution

$$p^{(1)}(x) = p^{(2)}(x) = \frac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}. \quad (29)$$

Performing the integrations in (16) and (17), we obtain

$$\omega^{(1)}(x) = \omega^{(2)}(x) = \frac{e^{-\frac{1}{4}x^2}}{2\sqrt{\pi}}. \quad (30)$$

Inserting this result into (22) and invoking (23) the problem is reduced to the calculation of the following integral

$$\begin{aligned} \mathcal{P}(q) &= \frac{1}{2\pi} \times 4 \times \frac{1}{(2\sqrt{\pi})^2} \int_{\sqrt{q}}^\infty dx \int_{-\sqrt{q}}^{\sqrt{q}} dy \frac{e^{-\frac{1}{4}\left(\frac{x-y}{2}\right)^2 - \frac{1}{4}\left(\frac{x+y}{2}\right)^2}}{\sqrt{(x^2 - q)(q - y^2)}} \\ &= \frac{1}{\pi^2} \int_{\sqrt{q}}^\infty dx \frac{e^{-\frac{1}{8}x^2}}{\sqrt{x^2 - q}} \int_0^{\sqrt{q}} dy \frac{e^{-\frac{1}{8}y^2}}{\sqrt{q - y^2}}, \end{aligned} \quad (31)$$

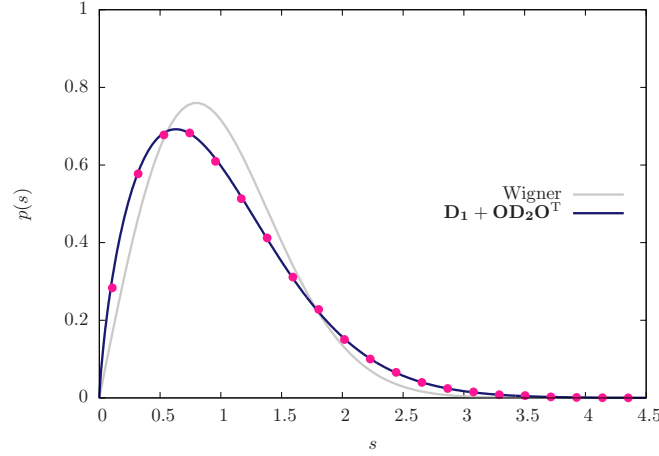


Figure 2. Spacing distribution for 2×2 random matrices formed as $\mathbf{D}_1 + \mathbf{O}\mathbf{D}_2\mathbf{O}^T$ (circles), where $\mathbf{D}_{1,2}$ are diagonal matrices whose diagonal entries are i.i.d. variables drawn from a standard Gaussian distribution, whereas $\mathbf{O}$ is a random Haar orthogonal matrix. The solid blue line refers to the analytical prediction (25) for this quantity specialized to the Gaussian case (see equation (33)). The grey solid line shows Wigner's surmise (A.2) for $\beta = 1$. Both the formula in equation (25), and the corresponding numerical results have been rescaled to have unit mean in order to be properly compared to Wigner's surmise. This can be easily achieved by mapping $p(s) \rightarrow \langle s \rangle p(\langle s \rangle s)$.

where in the last line we used the parity of the integrand. Performing the integrations we obtain

$$\mathcal{P}(q) = \frac{1}{4\pi} e^{-q/8} I_0\left(\frac{q}{16}\right) K_0\left(\frac{q}{16}\right), \quad (32)$$

where $I_0(x)$ and $K_0(x)$ are modified Bessel functions. Using (13) we get for the spacing distribution

$$p(s) = \frac{s}{2\pi} e^{-s^2/8} I_0\left(\frac{s^2}{16}\right) K_0\left(\frac{s^2}{16}\right), \quad (33)$$

which is correctly normalized, $\int_0^\infty ds p(s) = 1$. In Fig. 2 we show excellent agreement between equation(33) and the spacing distribution of numerically generated random matrices. The behavior as $s \rightarrow 0^+$ is indeed as predicted by (24)

$$p(s) \sim -\frac{1}{\pi} s \ln s, \quad (34)$$

since $\omega^{(1)}(0) = 1/2\sqrt{\pi}$ from (30).

4.2. Unitary case

We can rewrite (12) as

$$\mathcal{P}(q) = 2 \int_{-\infty}^{\infty} ds_1 ds_2 \int_0^{\pi/2} dt \sin(t) \cos(t) \delta(q - [s_1^2 + s_2^2 + 2s_1 s_2 \cos(2t)]) \omega^{(1)}(s_1) \omega^{(2)}(s_2). \quad (35)$$

Again, it is convenient to make the change of variables $s_1 = (x-y)/2$ and $s_2 = (x+y)/2$, yielding

$$\begin{aligned} \mathcal{P}(q) &= \int_{-\infty}^{\infty} dx dy \, \omega^{(1)}\left(\frac{x-y}{2}\right) \omega^{(2)}\left(\frac{x+y}{2}\right) \times \\ &\times \int_0^{\pi/2} dt \sin(t) \cos(t) \, \delta\left(q - \frac{1}{2}(x^2 + y^2) - \frac{1}{2}(x^2 - y^2) \cos(2t)\right). \end{aligned} \quad (36)$$

Setting $\cos(2t) = \xi$, one obtains

$$\begin{aligned} \mathcal{P}(q) &= \frac{1}{4} \int_{-\infty}^{\infty} dx dy \, \omega^{(1)}\left(\frac{x-y}{2}\right) \omega^{(2)}\left(\frac{x+y}{2}\right) \times \\ &\times \int_{-1}^1 d\xi \frac{\delta\left(\xi - \frac{q - (1/2)(x^2 + y^2)}{(1/2)(x^2 - y^2)}\right) \mathbb{1}\left(-1 \leq \frac{q - (1/2)(x^2 + y^2)}{(1/2)(x^2 - y^2)} \leq 1\right)}{(1/2)|x^2 - y^2|}. \end{aligned} \quad (37)$$

Resolving the constraints, we can express $\mathcal{P}(q)$ as the sum of four contributions

$$\mathcal{P}(q) = \frac{1}{4} \sum_{j=1}^4 I_j(q), \quad I_j(q) = \int_{X_j} dx \int_{Y_j} dy \, g(x, y), \quad (38)$$

where

$$g(x, y) = \frac{2\omega^{(1)}\left(\frac{x-y}{2}\right) \omega^{(2)}\left(\frac{x+y}{2}\right)}{|x^2 - y^2|} \quad (39)$$

and $X_1 = Y_2 = (-\infty, -\sqrt{q})$, $X_2 = X_3 = Y_1 = Y_4 = (-\sqrt{q}, \sqrt{q})$ and $X_4 = Y_3 = (\sqrt{q}, \infty)$. It is easy to see that if $\omega^{(1)}(x) = \omega^{(1)}(-x)$, then $I_1(q) = I_2(q)$ and $I_3(q) = I_4(q)$, and if $\omega^{(1)}(x) = \omega^{(2)}(x)$, then $I_2(q) = I_3(q)$.

Therefore, if $\omega^{(1)}(x) = \omega^{(2)}(x)$ is an even function, then $\mathcal{P}(q)$ can be simplified as

$$\mathcal{P}(q) = \int_{X_j} dx \int_{Y_j} dy \, g(x, y), \quad \text{any } j. \quad (40)$$

We shall mainly restrict to this case henceforth. In the unitary case, if $0 < \omega^{(1)}(0) < \infty$, we prove now that for $M = N = 2$ the level repulsion at zero is universally given by the following non-Wigner behavior

$$p(s) \sim \pi^2 [\omega^{(1)}(0)]^2 s, \quad s \rightarrow 0^+, \quad (41)$$

i.e. we observe a *linear* repulsion instead of the quadratic behavior one normally expects for unitarily invariant ensembles.

In order to prove (41), we start from (40) (combined with (13)) in the form

$$p(s) = 2s \int_s^\infty dx \int_{-s}^s dy \frac{2\omega^{(1)}\left(\frac{x-y}{2}\right) \omega^{(1)}\left(\frac{x+y}{2}\right)}{x^2 - y^2}. \quad (42)$$

For small s , the integral in y can be estimated as

$$\int_{-s}^s dy \frac{\omega^{(1)}\left(\frac{x-y}{2}\right) \omega^{(1)}\left(\frac{x+y}{2}\right)}{x^2 - y^2} \rightarrow \left[\omega^{(1)}\left(\frac{x}{2}\right)\right]^2 \int_{-s}^s \frac{dy}{x^2 - y^2} = 2 \left[\omega^{(1)}\left(\frac{x}{2}\right)\right]^2 \frac{\operatorname{arctanh}(s/x)}{x}. \quad (43)$$

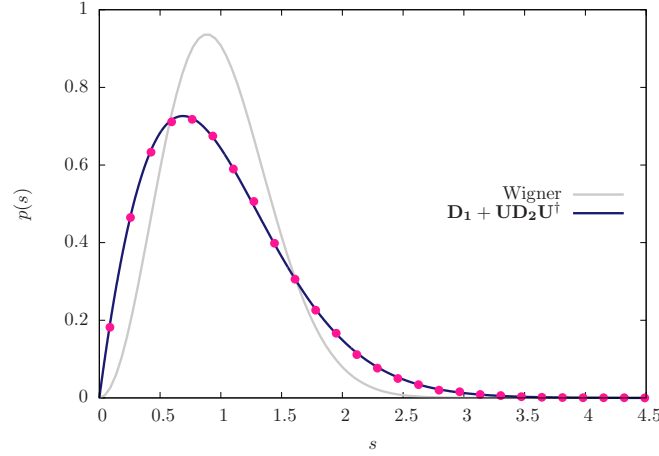


Figure 3. Spacing distribution for 2×2 random matrices formed as $\mathbf{D}_1 + \mathbf{U}\mathbf{D}_2\mathbf{U}^\dagger$ (circles), where $\mathbf{D}_{1,2}$ are diagonal matrices whose diagonal entries are i.i.d. variables drawn from a standard Gaussian distribution, whereas $\mathbf{U}$ is a random Haar unitary matrix. The solid blue line refers to the analytical prediction (42) for this quantity specialized to the Gaussian case (see equation (70)). The grey solid line shows Wigner's surmise (A.2) for $\beta = 2$. Both the formula in equation (42), and the corresponding numerical results have been rescaled to have unit mean in order to be properly compared to Wigner's surmise. This can be easily achieved by mapping $p(s) \rightarrow \langle s \rangle p(\langle s \rangle s)$.

Therefore

$$p(s) \sim 8s \int_s^\infty dx \left[\omega^{(1)}\left(\frac{x}{2}\right) \right]^2 \frac{\operatorname{arctanh}(s/x)}{x}. \quad (44)$$

For s close to zero we can estimate the integral contribution as

$$\int_s^\infty dx \left[\omega^{(1)}\left(\frac{x}{2}\right) \right]^2 \frac{\operatorname{arctanh}(s/x)}{x} \rightarrow [\omega^{(1)}(0)]^2 \int_s^1 dx \frac{\operatorname{arctanh}(s/x)}{x} \xrightarrow{s \rightarrow 0^+} \frac{\pi^2}{8} [\omega^{(1)}(0)]^2, \quad (45)$$

directly yielding (41). At the end of Sec. 5, we shall recover this behavior from an explicit calculation of the spacing distribution for Gaussian distributed eigenvalues, starting this time from the jpd of eigenvalues (4).

5. Rotational invariance and the jpd of eigenvalues for the complex hermitian (unitary) case

In this section, we first prove rotational invariance of our model, for simplicity in the unitary case.

The joint distribution of the entries of $\mathbf{S}$ can be written as

$$\mathcal{P}[\mathbf{S}] = \int \prod_{i=1}^M \left[d\mathbf{U}_i \prod_{j=1}^N d\lambda_j^{(i)} p^{(i)}(\lambda_j^{(i)}) \right] \delta \left(\mathbf{S} - \sum_{i=1}^M \mathbf{U}_i \mathbf{D}_i \mathbf{U}_i^\dagger \right), \quad (46)$$

where $\delta(\cdot)$ is a product of delta function, one for each independent entry of $\mathbf{S}$, and the first integrals are to be performed with a uniform Haar measure on the unitary group. We can now use the following integral representation of the matrix delta function

$$\delta(\mathbf{H}) = \frac{1}{(2\pi)^{N^2}} \int d\mathbf{T} e^{i\text{Tr}(\mathbf{T}\mathbf{H})}, \quad (47)$$

where $\mathbf{T}$ is a $N \times N$ hermitian matrix, to write

$$\mathcal{P}[\mathbf{S}] = \frac{1}{(2\pi)^{N^2}} \int d\mathbf{T} e^{-i\text{Tr}(\mathbf{T}\mathbf{S})} \int \prod_{i=1}^M \left[d\mathbf{U}_i \prod_{j=1}^N d\lambda_j^{(i)} p^{(i)}(\lambda_j^{(i)}) \right] e^{i\sum_{i=1}^M \text{Tr}(\mathbf{T}\mathbf{U}_i \mathbf{D}_i \mathbf{U}_i^\dagger)}. \quad (48)$$

Next, we can use the Harish-Chandra-Itzykson-Zuber (HCIZ) integral formula [17] to compute the $d\mathbf{U}_i$ integrals

$$\mathcal{P}[\mathbf{S}] \propto \int d\mathbf{T} e^{-i\text{Tr}(\mathbf{T}\mathbf{S})} \int \left[\prod_{j=1}^N \prod_{i=1}^M d\lambda_j^{(i)} p^{(i)}(\lambda_j^{(i)}) \right] \prod_{i=1}^M \frac{\det(e^{i\lambda_j^{(i)} t_k})_{j,k=1 \rightarrow N}}{\Delta(\boldsymbol{\lambda}^{(i)}) \Delta(\mathbf{t})}. \quad (49)$$

We can now diagonalize the hermitian matrix $\mathbf{T}$ by a unitary transformation $\mathbf{T} = \mathbf{W}\mathbf{Y}\mathbf{W}^\dagger$, with $\mathbf{Y}$ the diagonal matrix of eigenvalues $\mathbf{t} = \{t_1, \dots, t_N\}$ of $\mathbf{T}$. In a standard way, we get

$$\begin{aligned} \mathcal{P}[\mathbf{S}] &\propto \int \prod_{j=1}^N dt_j \Delta^2(\mathbf{t}) \int d\mathbf{W} e^{-i\text{Tr}(\mathbf{W}\mathbf{Y}\mathbf{W}^\dagger \mathbf{S})} \times \\ &\times \int \left[\prod_{j=1}^N \prod_{i=1}^M d\lambda_j^{(i)} p^{(i)}(\lambda_j^{(i)}) \right] \prod_{i=1}^M \frac{\det(e^{i\lambda_j^{(i)} t_k})_{j,k=1 \rightarrow N}}{\Delta(\boldsymbol{\lambda}^{(i)}) \Delta(\mathbf{t})}. \end{aligned} \quad (50)$$

We can now compute, for a given (arbitrary) unitary matrix $\mathbf{V}$,

$$\begin{aligned} \mathcal{P}[\mathbf{V}\mathbf{S}\mathbf{V}^\dagger] &\propto \int \prod_{j=1}^N dt_j \Delta^2(\mathbf{t}) \int d\mathbf{W} e^{-i\text{Tr}(\mathbf{W}\mathbf{Y}\mathbf{W}^\dagger \mathbf{V}\mathbf{S}\mathbf{V}^\dagger)} \times \\ &\times \int \left[\prod_{j=1}^N \prod_{i=1}^M d\lambda_j^{(i)} p^{(i)}(\lambda_j^{(i)}) \right] \prod_{i=1}^M \frac{\det(e^{i\lambda_j^{(i)} t_k})_{j,k=1 \rightarrow N}}{\Delta(\boldsymbol{\lambda}^{(i)}) \Delta(\mathbf{t})}. \end{aligned} \quad (51)$$

Using the cyclic property of the trace, renaming $\mathbf{V}^\dagger \mathbf{W} \rightarrow \mathbf{W}$ and using invariance of the Haar measure, we indeed obtain that (51) is equivalent to (50), proving that $\mathcal{P}[\mathbf{S}] = \mathcal{P}[\mathbf{V}\mathbf{S}\mathbf{V}^\dagger]$.

Using now the change of variables

$$\mathcal{P}[\mathbf{S}] dS_{11} \cdots dS_{NN} = \mathcal{P}(\boldsymbol{\nu}) d\nu_1 \cdots d\nu_N \prod_{i,j} dV_{ij}, \quad (52)$$

where $\boldsymbol{\nu} = \{\nu_1, \dots, \nu_N\}$ are the eigenvalues of $\mathbf{S}$, $\mathcal{P}(\boldsymbol{\nu})$ their jpd and V_{ij} the independent components of the eigenvectors of $\mathbf{S}$, we obtain for the jpd of eigenvalues of $\mathbf{S}$

$$\begin{aligned} \mathcal{P}(\boldsymbol{\nu}) &\propto \Delta^2(\boldsymbol{\nu}) \int \prod_{j=1}^N dt_j \Delta^2(\mathbf{t}) \int d\mathbf{V} d\mathbf{W} e^{-i\text{Tr}(\mathbf{W}\mathbf{Y}\mathbf{W}^\dagger \mathbf{V}\mathbf{N}\mathbf{V}^\dagger)} \times \\ &\times \int \left[\prod_{j=1}^N \prod_{i=1}^M d\lambda_j^{(i)} p^{(i)}(\lambda_j^{(i)}) \right] \prod_{i=1}^M \frac{\det(e^{i\lambda_j^{(i)} t_k})_{j,k=1 \rightarrow N}}{\Delta(\boldsymbol{\lambda}^{(i)}) \Delta(\mathbf{t})}, \end{aligned} \quad (53)$$

where $\mathbf{N} = \text{diag}(\nu_1, \dots, \nu_N)$. Using again the cyclic property of the trace, and the invariance of the Haar measure, we can perform another HCIZ integral to get to

$$\begin{aligned} \mathcal{P}(\boldsymbol{\nu}) &\propto \Delta^2(\boldsymbol{\nu}) \int \prod_{j=1}^N dt_j \Delta^2(\mathbf{t}) \frac{\det(e^{-i\nu_j t_k})_{j,k=1 \rightarrow N}}{\Delta(\boldsymbol{\nu}) \Delta(\mathbf{t})} \times \\ &\times \int \left[\prod_{j=1}^N \prod_{i=1}^M d\lambda_j^{(i)} p^{(i)}(\lambda_j^{(i)}) \right] \prod_{i=1}^M \frac{\det(e^{i\lambda_j^{(i)} t_k})_{j,k=1 \rightarrow N}}{\Delta(\boldsymbol{\lambda}^{(i)}) \Delta(\mathbf{t})}, \end{aligned} \quad (54)$$

which after simplifications reduces to the $N(M+1)$ -fold integral

$$\begin{aligned} \mathcal{P}(\boldsymbol{\nu}) &\propto \Delta(\boldsymbol{\nu}) \int \prod_{j=1}^N dt_j \left[\prod_{j=1}^N \prod_{i=1}^M d\lambda_j^{(i)} p^{(i)}(\lambda_j^{(i)}) \right] \times \\ &\times \frac{\det(e^{-i\nu_j t_k})_{j,k=1 \rightarrow N}}{[\Delta(\mathbf{t})]^{M-1}} \prod_{i=1}^M \frac{\det(e^{i\lambda_j^{(i)} t_k})_{j,k=1 \rightarrow N}}{\Delta(\boldsymbol{\lambda}^{(i)})}, \end{aligned} \quad (55)$$

as in (4).

In the case where the distribution of eigenvalues is the same for all summands, $p^{(i)}(x) \equiv p(x)$, $\forall i$, Eq. (55) simplifies to

$$\mathcal{P}(\boldsymbol{\nu}) \propto \Delta(\boldsymbol{\nu}) \int \prod_{j=1}^N dt_j \frac{\det(e^{-i\nu_j t_k})_{j,k=1 \rightarrow N}}{[\Delta(\mathbf{t})]^{M-1}} \left[\int \prod_{j=1}^N d\lambda_j p(\lambda_j) \frac{\det(e^{i\lambda_j t_k})_{j,k=1 \rightarrow N}}{\Delta(\boldsymbol{\lambda})} \right]^M. \quad (56)$$

Let us consider some limiting cases of this jpd. For $M = 1$ (no sums), we expect that the jpd of eigenvalues $\boldsymbol{\nu}$ reproduces the jpd of eigenvalues $\boldsymbol{\lambda}$, i.e.

$$\mathcal{P}(\boldsymbol{\nu}) = \prod_{j=1}^N p^{(1)}(\nu_j), \quad (57)$$

as the ensemble of matrices $\mathbf{S}$ just contains in this case randomly rotated (and therefore similar) diagonal matrices with identical distribution of elements. Setting $M = 1$ in (55) or (56), we have first to evaluate

$$\int_{-\infty}^{\infty} \prod_{j=1}^N dt_j \det(e^{-i\nu_j t_k})_{j,k=1 \rightarrow N} \det(e^{i\lambda_j^{(1)} t_k})_{j,k=1 \rightarrow N} \propto \det \left(\delta(\lambda_j^{(1)} - \nu_k) \right)_{j,k=1 \rightarrow N}, \quad (58)$$

where we have used the standard Andréief identity [18]. Inserting (58) into (55) and expanding the determinant of delta functions, we precisely obtain (57) after making a suitable number of sign changes in the denominator $\Delta(\boldsymbol{\lambda}^{(1)})$ (with $\lambda_j^{(1)}$ replaced by ν_k).

5.1. Spacing distribution: complex hermitian (unitary) case with $N = M = 2$ and Gaussian law

Let us now stress that the denominator $[\Delta(\mathbf{t})]^{M-1}$ in (55) is indeed responsible for $M \geq 2$ for the non-trivial nature of our ensemble. Let us consider e.g. the case $N = M = 2$ with a standard Gaussian distribution $p^{(i)}(x) = e^{-x^2/2}/\sqrt{2\pi}$, $\forall i$. We should then recover from (55) the results in Section 4.2, specialized to the Gaussian law.

We get for $N = 2$ (ignoring prefactors)

$$\begin{aligned} \mathcal{P}(\nu_1, \nu_2) &\propto (\nu_2 - \nu_1) \int_{-\infty}^{\infty} dt_1 dt_2 d\lambda_1 d\lambda_2 d\ell_1 d\ell_2 \times \\ &\times \exp\left(-\frac{\lambda_1^2}{2} - \frac{\lambda_2^2}{2} - \frac{\ell_1^2}{2} - \frac{\ell_2^2}{2}\right) \frac{\psi(t_1, t_2, \lambda_1, \lambda_2, \ell_1, \ell_2, \nu_1, \nu_2)}{(t_2 - t_1)(\lambda_2 - \lambda_1)(\ell_2 - \ell_1)}, \end{aligned} \quad (59)$$

where

$$\begin{aligned} \psi(t_1, t_2, \lambda_1, \lambda_2, \ell_1, \ell_2, \nu_1, \nu_2) &= [e^{-i(\nu_1 t_1 + \nu_2 t_2)} - e^{-i(\nu_1 t_2 + \nu_2 t_1)}] [e^{i(\lambda_1 t_1 + \lambda_2 t_2)} - e^{i(\lambda_1 t_2 + \lambda_2 t_1)}] \times \\ &\times [e^{i(\ell_1 t_1 + \ell_2 t_2)} - e^{i(\ell_1 t_2 + \ell_2 t_1)}]. \end{aligned} \quad (60)$$

After lengthy algebra, we obtain

$$\mathcal{P}(\nu_1, \nu_2) \propto (\nu_2 - \nu_1) I(\nu_1, \nu_2), \quad (61)$$

where

$$\begin{aligned} I(\nu_1, \nu_2) &= \int_{-\infty}^{\infty} \frac{dt_1 dt_2 (e^{-i(\nu_1 t_1 + \nu_2 t_2)} - e^{-i(\nu_1 t_2 + \nu_2 t_1)})}{t_2 - t_1} \times \\ &\times \left[\int_{-\infty}^{\infty} \frac{dx dy}{y - x} \exp\left(-\frac{1}{2}(x^2 + y^2)\right) (e^{i(xt_1 + yt_2)} - e^{i(xt_2 + yt_1)}) \right]^2. \end{aligned} \quad (62)$$

The integral in square brackets can be performed making the change of variables $\tau = (x + y)/2$ and $\zeta = (x - y)/2$ yielding

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{dx dy}{y - x} \exp\left(-\frac{1}{2}(x^2 + y^2)\right) (e^{i(xt_1 + yt_2)} - e^{i(xt_2 + yt_1)}) &= \\ &= -2i\pi^{3/2} e^{-(1/4)(t_1 + t_2)^2} \operatorname{erf}\left(\frac{t_1 - t_2}{2}\right), \end{aligned} \quad (63)$$

where $\operatorname{erf}(z) = (2/\sqrt{\pi}) \int_0^z e^{-t^2} dt$ is the error function. It is now convenient to introduce the distribution $p(s)$ of the spacing $s = \nu_2 - \nu_1$ between the two eigenvalues

$$p(s) = 2 \int_{-\infty}^{\infty} d\nu_1 d\nu_2 \mathcal{P}(\nu_1, \nu_2) \delta(s - (\nu_2 - \nu_1)) = 2 \int_{-\infty}^{\infty} d\nu \mathcal{P}(\nu, \nu + s). \quad (64)$$

Performing the remaining integrals in (62), we are left with

$$p(s) \propto s J_2(s), \quad (65)$$

where

$$J_2(s) = \int_{-\infty}^{\infty} dt \frac{\sin(st)}{t} (\operatorname{erf}(t))^2 = 2 \int_0^{\infty} dt \frac{\sin(st)}{t} (\operatorname{erf}(t))^2. \quad (66)$$

In (66), the term $(\operatorname{erf}(t))^2$ descends from the double integral evaluation in (63) which needs to be squared (see (62)). The term $\sin(st)$ instead comes from the phase difference in (62) (first line).

The integral $J_2(s)$ can be written in the alternative and more convenient form

$$J_2(s) = 4 \int_0^{\pi/4} d\theta \operatorname{erfc}\left(\frac{s}{2} \cos \theta\right), \quad (67)$$

where $\operatorname{erfc}(z) = 1 - \operatorname{erf}(z)$ is the complementary error function. To go from (66) to (67), one first writes $(\operatorname{erf}(t))^2 = (4/\pi) \int_{(0,t)^2} dz_1 dz_2 \exp(-z_1^2 - z_2^2)$, then performs the change of variables $z_{1,2} = t\zeta_{1,2}$. The integral in t can be performed first, and the remaining integrals in $\zeta_{1,2}$ from 0 to 1 can be solved in polar coordinates, leaving eventually the angular integral in (67) which cannot be evaluated in closed form.

This representation allows to fix the normalization of $p(s)$ using

$$\int_0^\infty ds \, s \operatorname{erfc}\left(\frac{s}{2} \cos \theta\right) = \frac{1}{(\cos \theta)^2} , \quad (68)$$

$$\int_0^{\pi/4} d\theta \frac{1}{(\cos \theta)^2} = 1 . \quad (69)$$

Eventually we obtain precisely

$$p(s) = s \int_0^{\pi/4} d\theta \operatorname{erfc}\left(\frac{s}{2} \cos \theta\right) , \quad (70)$$

(normalized to 1) whose behavior for $s \rightarrow 0^+$ is $p(s) \sim (\pi/4)s + \dots$, in agreement with the general result (41). In Fig. 3 we show a perfect agreement between numerically generated matrices and the spacing distribution (70).

5.2. Spacing distribution: complex hermitian (unitary) case with $N = 2$, M generic and Gaussian law

Using the results of previous subsection, we can now tackle another case, namely the complex hermitian (unitary) case with $N = 2$ and $M > 2$. Even with the smallest possible size ($N = 2$), this case cannot be efficiently dealt with in the framework of Section 4.2, based on the matrix definition (3). However, we can still exploit the exact jpd (56) and the integral (62) (with the exponent of the square bracket replaced with M and $t_2 - t_1$ replaced by $(t_2 - t_1)^{M-1}$) to get for the spacing distribution

$$p(s) = K_M s J_M(s) , \quad (71)$$

where

$$J_M(s) = \int_0^\infty dt \frac{\sin(st)}{t^{M-1}} [\operatorname{erf}(t)]^M . \quad (72)$$

Note that for $M = 2$, (71) reproduces (65) as it should. For $M > 2$, the algebraic manipulations yielding from (66) to (67) do not seem to work, therefore the normalization constant K_M must be fixed case by case. However, it is quite easy to find out that for $M \geq 4$ the behavior for small spacings is *quadratic*

$$p(s) \sim K_M \omega_M s^2, \quad s \rightarrow 0^+ , \quad (73)$$

with

$$\omega_M = \int_0^\infty dt \frac{[\operatorname{erf}(t)]^M}{t^{M-2}} . \quad (74)$$

In some sense, for $M \geq 4$ we recover a “Wigner-like” behavior, even though the details of the spacing distribution are clearly different from the Wigner’s surmise (A.2) for

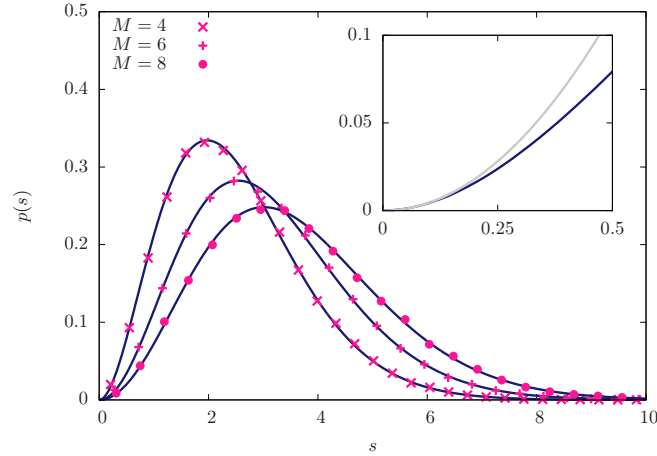


Figure 4. Comparison between equation (71) and the distribution of eigenvalue spacings obtained numerically from the addition of $M = 4, 6, 8$ unitary 2×2 random matrices (see equation (3)). The inset shows the comparison between the exact result of equation (71) (dark blue line) and its asymptotic behavior for small s (73) for the case $M = 4$.

$\beta = 2$. In Sec. 6.3, we will indeed see that for $M \gg 1$ the spacing distribution (for *any* distribution of eigenvalues with finite variance σ^2) attains a scaling form $p(s) \rightarrow \frac{1}{\sigma\sqrt{M}} \Phi_{\mathbf{U}}\left(\frac{s}{\sigma\sqrt{M}}\right)$, where the scaling function $\Phi_{\mathbf{U}}(x)$ (see (107)) is different from (A.2).

In Fig. 4, we include plots of the spacing distributions for different M (this time, without adjusting the average $\langle s \rangle$ to 1), together with numerical simulations.

The case $M = 3$ is instead different (note that for $M = 3$, the constant ω_3 in (74) would be divergent). The behavior for small spacing is quite exotic,

$$p(s) \sim -K_3 s^2 \ln s, \quad s \rightarrow 0^+, \quad (75)$$

and is determined by the small s behavior of the integral $J_3(s)$ from (72), $J_3(s) \sim -s \ln s$ for $s \rightarrow 0^+$.

6. Spacing between the $N = 2$ eigenvalues for $M \gg 1$: interpretation in terms of sums of random vectors

In this Section, we discuss the asymptotic properties of the spacing distribution $p(s)$ of the invariant sum (3) for $N = 2$ and $M \gg 1$. Hereafter we will assume that the eigenvalues of the diagonal matrices $\mathbf{D}_i$ are all drawn from the same distribution with finite variance σ^2 . We are able to show that $p(s)$ attains for $M \gg 1$ a scaling form

$$p(s) = \frac{1}{\sigma\sqrt{M}} \Phi_{\mathbf{O}}\left(\frac{s}{\sigma\sqrt{M}}\right), \quad p(s) = \frac{1}{\sigma\sqrt{M}} \Phi_{\mathbf{U}}\left(\frac{s}{\sigma\sqrt{M}}\right) \quad (76)$$

for the orthogonal and unitary cases respectively. The scaling functions $\Phi_{\mathbf{O}}(x)$ and $\Phi_{\mathbf{U}}(x)$ can be calculated exactly using a quite elegant interpretation of the spacing distribution $p(s)$ in terms of sums of random vectors in a plane (orthogonal case) or in

the full three-dimensional space (unitary case). We start by recalling the statement of the Central Limit Theorem for the multidimensional (vectorial) case.

6.1. Multidimensional Central Limit Theorem

Let $\{\mathbf{x}_1, \dots, \mathbf{x}_M\}$ be a collection of M independent and identically distributed^{||} vectors in $\mathbb{R}^k$ with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ (amongst the individual components of the vectors). $\boldsymbol{\Sigma}$ is therefore a $k \times k$ symmetric and positive-semidefinite matrix.

Let

$$\mathbf{x}_i = \begin{pmatrix} x_{i(1)} \\ \vdots \\ x_{i(k)} \end{pmatrix} \quad (77)$$

be the i th vector, and the sum $\mathbf{s}_M = \sum_{i=1}^M \mathbf{x}_i$. Then the multidimensional Central Limit Theorem states that the normalized sum $\mathbf{s} = (\mathbf{s}_M - M\boldsymbol{\mu})/\sqrt{M}$ converges for $M \rightarrow \infty$ to a multivariate normal (Gaussian) distribution

$$\mathbf{s} \xrightarrow{D} \mathcal{N}_k(0, \boldsymbol{\Sigma}) . \quad (78)$$

Explicitly, the probability density of a multivariate Gaussian variable $\mathbf{s} \sim \mathcal{N}_k(0, \boldsymbol{\Sigma})$ reads

$$\mathcal{P}(\mathbf{s}) = \frac{1}{(2\pi)^{k/2} \sqrt{\det \boldsymbol{\Sigma}}} e^{-\frac{1}{2} \mathbf{s}^T \boldsymbol{\Sigma}^{-1} \mathbf{s}} . \quad (79)$$

We will be mostly interested in the cases $k = 2$ and $k = 3$. A natural question is then: what is the distribution of the *length* (the modulus $|\mathbf{s}|$) of the limiting vector sum $\mathbf{s}$ distributed as in (79)? Clearly, an enormous simplification in (79) occurs if the covariance matrix of individual vector components is a multiple of the identity, allowing to use a spherical coordinate transformation. We will compute this distribution for $k = 2$ and $k = 3$ in due course, after explaining why this machinery turns out to be very useful for our RMT problem.

6.2. Spacing of the matrix sum $\mathbf{S}$ as the sum of 2D or 3D random vectors

Three-dimensional vectors $\mathbf{x} = (x_1, x_2, x_3)$ can be mapped onto 2×2 matrices $\mathbf{X} = \mathbf{x} \cdot \boldsymbol{\sigma} = x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3$, *i.e.* linear combinations of the Pauli matrices

$$\boldsymbol{\sigma}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} , \quad \boldsymbol{\sigma}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} , \quad \boldsymbol{\sigma}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} . \quad (80)$$

^{||} The concepts of independence and identical distribution apply to any two different vectors. Within a single vector, its components may well be correlated or non-identically distributed.

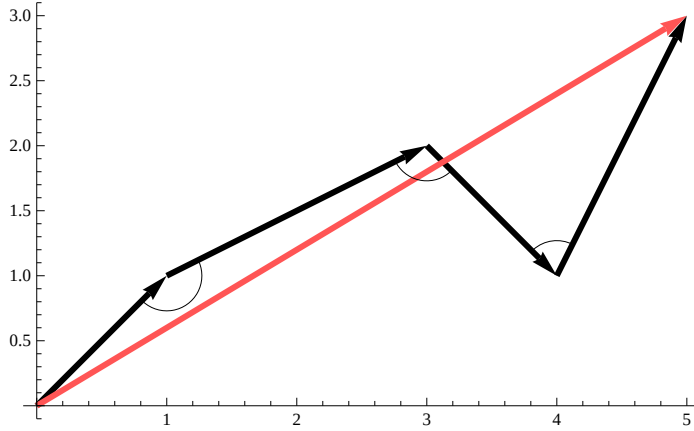


Figure 5. Sketch of the sum of randomly oriented planar vectors.

Explicitly, we have

$$\mathbf{X} = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (81)$$

Note in particular that a vector aligned with the z -axis ($x_1 = x_2 = 0$) corresponds to a *diagonal* matrix $\mathbf{X}$. The norm of a vector is then given by $|\mathbf{x}|^2 = -\det \mathbf{X}$, while rotations of a vector $\mathbf{x} \rightarrow \mathbf{x}' = \mathbf{R}\mathbf{x}$ by an angle 2ϕ about the axis determined by a unit vector $\mathbf{n}$ are given by the formula

$$\mathbf{X} \rightarrow \mathbf{X}' = \mathbf{U}\mathbf{X}\mathbf{U}^\dagger \quad (82)$$

where $\mathbf{X}' = \mathbf{x}' \cdot \boldsymbol{\sigma}$ and $\mathbf{U} = e^{-i\phi\mathbf{n} \cdot \boldsymbol{\sigma}} = \cos(\phi)\boldsymbol{\sigma}_0 - i\sin(\phi)\boldsymbol{\sigma} \cdot \mathbf{n}$ is a $SU(2)$ matrix ($\boldsymbol{\sigma}_0$ denotes the 2×2 identity matrix). Explicitly we get

$$\mathbf{U} = \begin{pmatrix} \cos(\phi) - in_3 \sin(\phi) & (-in_1 - n_2) \sin(\phi) \\ (-in_1 + n_2) \sin(\phi) & \cos(\phi) + in_3 \sin(\phi) \end{pmatrix}. \quad (83)$$

When rotations about the second axis $\mathbf{n} = (0, -1, 0)$ are considered, the matrix $\mathbf{U}$ takes the form of an orthogonal matrix $\mathbf{U} = \mathbf{O}$

$$\mathbf{O} = \cos(\phi)\boldsymbol{\sigma}_0 + i\sin(\phi)\boldsymbol{\sigma}_2 = \begin{pmatrix} \cos(\phi) & \sin(\phi) \\ -\sin(\phi) & \cos(\phi) \end{pmatrix}. \quad (84)$$

Let us apply this mapping between vectors and 2×2 matrices to analyze the statistical properties of the invariant sum $\mathbf{S}$, starting with the orthogonal case.

6.2.1. Orthogonal case In this case, we have

$$\mathbf{S} = \sum_{i=1}^M \mathbf{O}_i \mathbf{D}_i \mathbf{O}_i^T \quad (85)$$

where

$$\mathbf{D}_i = \begin{pmatrix} \lambda_1^{(i)} & 0 \\ 0 & \lambda_2^{(i)} \end{pmatrix}, \quad \mathbf{O}_i = \begin{pmatrix} \cos(\phi_i) & \sin(\phi_i) \\ -\sin(\phi_i) & \cos(\phi_i) \end{pmatrix}, \quad (86)$$

the ϕ_i 's being independent random variables uniformly distributed over $[0, \pi/2]$, and the $\lambda_j^{(i)}$'s are i.i.d. random variables for $i = 1, \dots, M$ and $j = 1, 2$. In (9), we used the identification $\phi_{i=1} \equiv t$. The diagonal matrices $\mathbf{D}_i$ can be written as

$$\mathbf{D}_i = \frac{t_i}{2} \boldsymbol{\sigma}_0 + \frac{s_i}{2} \boldsymbol{\sigma}_3, \quad (87)$$

where $t_i = \lambda_1^{(i)} + \lambda_2^{(i)}$ is the trace of $\mathbf{D}_i$, and $s_i = \lambda_1^{(i)} - \lambda_2^{(i)}$ is the spacing between the unsorted eigenvalues of $\mathbf{D}_i$ (so $s_i \leq 0$). From equation (87) we see that the invariant sum (85) takes the form

$$\mathbf{S} = \frac{1}{2} \sum_{i=1}^M t_i \boldsymbol{\sigma}_0 + \frac{1}{2} \sum_{i=1}^M \mathbf{O}_i (s_i \boldsymbol{\sigma}_3) \mathbf{O}_i^T. \quad (88)$$

The first term shows that the trace of the sum is a sum of traces of individual terms, whereas the second term can be interpreted as a sum of planar vectors embedded in the plane (x, z) . Indeed, each term in the sum

$$\mathbf{O}_i (s_i \boldsymbol{\sigma}_3) \mathbf{O}_i^T = \mathbf{x}'_i \cdot \boldsymbol{\sigma} = -\sin(2\phi_i) s_i \boldsymbol{\sigma}_1 + \cos(2\phi_i) s_i \boldsymbol{\sigma}_3 \quad (89)$$

represents a vector $\mathbf{x}'_i$ of length s_i obtained rotating the vector $\mathbf{x}_i = (0, 0, s_i)$ around the second axis by an angle $2\phi_i$. The rotated vector

$$\mathbf{x}'_i = (-\sin(2\phi_i) s_i, 0, \cos(2\phi_i) s_i) \quad (90)$$

remains in the plane (x, z) and so does the sum $\mathbf{s} = \sum_i \mathbf{x}'_i$, corresponding to the second term in (88), $\sum_{i=1}^M \mathbf{O}_i (s_i \boldsymbol{\sigma}_3) \mathbf{O}_i^T$. This vector $\mathbf{s}$ can be rotated back and aligned to the third axis z by a rotation about the second axis. But recall that a vector aligned to the axis z corresponds to a 2×2 diagonal matrix, from (81). Hence, this rotation of the vector $\mathbf{s}$ corresponds, in the matrix language, to the diagonalization of $\sum_{i=1}^M \mathbf{O}_i (s_i \boldsymbol{\sigma}_3) \mathbf{O}_i^T$. Using (88), one concludes that the matrix $\mathbf{S}$ itself becomes diagonal, i.e.

$$\left(\sum_{i=1}^M \mathbf{x}'_i \right) \cdot \boldsymbol{\sigma} = \mathbf{O} (s \boldsymbol{\sigma}_3) \mathbf{O}^T, \quad (91)$$

where $\mathbf{O} = \cos(\phi) \boldsymbol{\sigma}_0 + i \sin(\phi) \boldsymbol{\sigma}_2$ is the orthogonal matrix that diagonalizes $\mathbf{S}$, and 2ϕ is the angle between $\mathbf{s}$ and the z -axis. The eigenvalues of $\mathbf{S}$, combining (88) and (91), can be written as $\nu_1 = t + s$ and $\nu_2 = t - s$, implying that s is indeed the spacing of the invariant sum $\mathbf{S}$. Since rotation does not change the length $s = |\mathbf{s}|$ of the vector, we see from (88) that the spacing (between the *largest* and the *smallest*, i.e. $s > 0$) is given by length of the sum $\mathbf{s} = \sum_i \mathbf{x}'_i$ of M identically distributed random vectors with random directions in the plane (x, z) . For $M \gg 1$, we can then apply the multidimensional Central Limit Theorem as in (78): all we need is the average vector $\boldsymbol{\mu}$ and the covariance $\boldsymbol{\Sigma}$ between the components of each summand.

We have from (90)

$$\langle \sin(2\phi_i) s_i \rangle = \langle \cos(2\phi_i) s_i \rangle = 0 , \quad (92)$$

$$\langle (\sin(2\phi_i) s_i)^2 \rangle = \langle (\cos(2\phi_i) s_i)^2 \rangle = \frac{1}{2} \times 2\sigma^2 = \sigma^2 , \quad (93)$$

$$\langle \sin(2\phi_i) \cos(2\phi_i) s_i^2 \rangle = 0 , \quad (94)$$

where, since the angles and the components s_i are independent of each other, the average $\langle (\cdots) \rangle$ factorizes into the product of the average over a flat distribution of the angles over $[0, \pi/2]$, and the second moment of the individual spacing $\langle (s_i)^2 \rangle = 2\sigma^2$, if σ^2 is the variance of the individual eigenvalues $\lambda_{(1,2)}^{(i)}$. Note that we have used the fact that $\langle s_i \rangle = 0$, as the s_i can be ≤ 0 (contrary to the total spacing $s > 0$) and we recall the assumption that $\lambda_{1,2}^{(i)}$ have been both drawn from the same distribution with finite variance σ^2 . Hence, the covariance matrix Σ is in this case a multiple of the identity,

$$\Sigma = \sigma^2 \sigma_0 . \quad (95)$$

Taking into account the scaling factor $\sqrt{M}$ that appears on the left hand side of (78), we conclude that the distribution of the *length* of the vector $s = |\mathbf{s}|$ (i.e. the spacing of the sum $\mathbf{S}$) can be computed from (79) as

$$p(s) \propto \int_{(-\infty, \infty)^2} ds_1 ds_2 e^{-\frac{s_1^2 + s_2^2}{2M\sigma^2}} \delta\left(s - \sqrt{s_1^2 + s_2^2}\right) , \quad (96)$$

which can be easily computed in polar coordinates, yielding eventually the scaling form for $M \gg 1$

$$p(s) = \frac{1}{\sigma\sqrt{M}} \Phi_{\mathbf{O}}\left(\frac{s}{\sigma\sqrt{M}}\right) , \quad \Phi_{\mathbf{O}}(x) = x e^{-\frac{1}{2}x^2} . \quad (97)$$

Note that the scaling function $\Phi_{\mathbf{O}}(x)$ has precisely the form of the Wigner's surmise $p_{WS}^{(\beta=1)}(x)$ for $\beta = 1$ (see (A.2)), giving the spacing distribution of Gaussian matrices with orthogonal symmetry (see Figure 6 for a numerical validation). Therefore, in the orthogonal case the spacing distributions in the limiting situations $N \gg 1, M = 2$ (on unfolded eigenvalues) and $N = 2, M \gg 1$ are the same.

6.3. Unitary case

We can repeat the same argument for the unitary case

$$\mathbf{S} = \sum_{i=1}^M \mathbf{U}_i \mathbf{D}_i \mathbf{U}_i^\dagger . \quad (98)$$

The main difference with respect to the previous case is that the rotations

$$\mathbf{U}_i(s_i \boldsymbol{\sigma}_3) \mathbf{U}_i^\dagger = \mathbf{x}'_i \cdot \boldsymbol{\sigma} \quad (99)$$

produce vectors $\mathbf{x}'$ which have random directions not in two but in three dimensional space,

$$\mathbf{x}' = (s_i \sin(2t) \cos(\theta + \phi_1 + \phi_2), s_i \sin(2t) \sin(\theta + \phi_1 + \phi_2), s_i \cos(2t)) , \quad (100)$$

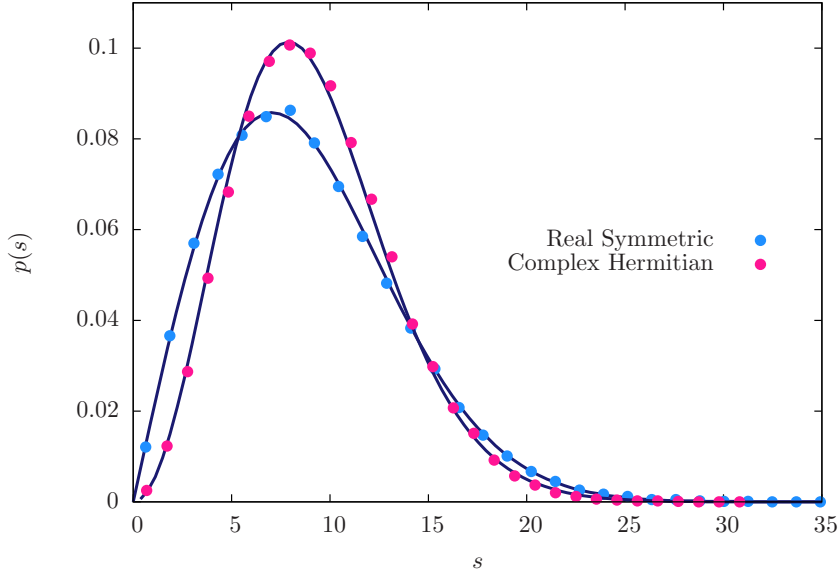


Figure 6. Spacing distribution for 2×2 real symmetric matrices formed as in equation (85) and 2×2 complex Hermitian matrices formed as in equation (98), where entries in the diagonal matrices $\mathbf{D}_i$ are i.i.d. random variables drawn from a standard Gaussian distribution. Numerical results (dots) refer to the sum of $M = 50$ matrices, whereas the blue solid lines represent the analytical predictions of equation (97) (orthogonal case) and (107) (unitary case) with $\sigma = 1$.

where we have used the parametrization (8) for a general 2×2 unitary matrix. We then have

$$\langle s_i \sin(2t) \cos(\theta + \phi_1 + \phi_2) \rangle = \langle s_i \sin(2t) \sin(\theta + \phi_1 + \phi_2) \rangle = \langle \cos(2t) \rangle = 0, \quad (101)$$

$$\langle (s_i \sin(2t) \cos(\theta + \phi_1 + \phi_2))^2 \rangle = \langle (s_i \sin(2t) \sin(\theta + \phi_1 + \phi_2))^2 \rangle = \frac{1}{4} \times 2\sigma^2 = \frac{1}{2}\sigma^2, \quad (102)$$

$$\langle (s_i \cos(2t))^2 \rangle = \frac{1}{2} \times 2\sigma^2 = \sigma^2, \quad (103)$$

$$\begin{aligned} \langle s_i^2 \sin(2t)^2 \cos(\theta + \phi_1 + \phi_2) \sin(\theta + \phi_1 + \phi_2) \rangle &= \langle s_i^2 \sin(2t) \cos(2t) \cos(\theta + \phi_1 + \phi_2) \rangle = \\ &= \langle s_i^2 \sin(2t) \cos(2t) \sin(\theta + \phi_1 + \phi_2) \rangle = 0. \end{aligned} \quad (104)$$

Hence, in this case the covariance matrix is still diagonal, but is no longer a multiple of the identity

$$\Sigma = \sigma^2 \text{diag} \left(\frac{1}{2}, \frac{1}{2}, 1 \right), \quad (105)$$

implying that the spacing distribution (invoking again (78) and (79)) reads in this case

$$p(s) \propto \int_{(-\infty, \infty)^3} ds_1 ds_2 ds_3 \exp \left[-\frac{1}{2M\sigma^2} (2s_1^2 + 2s_2^2 + s_3^2) \right] \delta \left(s - \sqrt{s_1^2 + s_2^2 + s_3^2} \right). \quad (106)$$

The integral can be still performed in spherical coordinates, and again a scaling form (valid for $M \gg 1$) is attained

$$p(s) = \frac{1}{\sigma\sqrt{M}} \Phi_{\mathbf{U}} \left(\frac{s}{\sigma\sqrt{M}} \right), \quad \Phi_{\mathbf{U}}(x) = 2x e^{-x^2} \text{erfi} \left(\frac{x}{\sqrt{2}} \right), \quad (107)$$

where $\operatorname{erfi}(z) = -i \operatorname{erf}(iz)$. The scaling function $\Phi_{\mathbf{U}}(x)$ is correctly normalized $\int_0^\infty dx \Phi_{\mathbf{U}}(x) = 1$, and for $x \rightarrow 0^+$ it goes like $\Phi_{\mathbf{U}}(x) \sim 2\sqrt{2/\pi}x^2$. The quadratic behavior is in agreement with the finite- M formulae (71) and (72), valid for $M \geq 4$. However, it is not easy to derive (107) directly from (72). The scaling function is in this case significantly different from the Wigner’s surmise for $\beta = 2$ (see (A.2) below). In Figure 6 a numerical validation of equation (107) is shown.

7. Conclusions

In summary, using a simple “rotate and sum” prescription, we constructed and explicitly solved an ensemble of random matrices with rather peculiar features. It is by construction invariant with respect to classical symmetry groups (orthogonal and unitary) and yet its eigenvalue repulsion is not precisely Vandermondean. Furthermore, by carefully choosing the distribution of independent diagonal entries $p^{(i)}(x)$ (e.g. a semicircle law), the ensemble for large matrix size N (even at fixed number of addenda $M \geq 2$) reproduces *both* macroscopic (level density) *and* microscopic (spacing distribution and two-points correlation functions) spectral statistics of the standard Gaussian ensemble. In this sense, classical RMT universality is restored only asymptotically, and we conjecture that microscopic features (such as the spacing distribution and the two-point correlation function) will be universally given by the corresponding GOE and GUE results for $N \rightarrow \infty$, irrespective of the marginal distribution $p^{(i)}(x)$ of the diagonal elements and the number M of summands.

We analyzed in detail the case $N = M = 2$ and found in full generality an unexpected behavior of the spacing distribution $p(s) \approx -s \ln s$ (orthogonal case) and $p(s) \approx s$ (unitary case) at small spacings, markedly different from the standard Wigner-Dyson or Poisson laws for classical invariant ensembles and uncorrelated variables, respectively. For $N = 2$ and $M \geq 3$, different behaviors for the spacings also appear (see Section 5.2), while for $N = 2$ and $M \gg 1$ the spacing distribution $p(s)$ attains scaling forms that can be understood in terms of the multidimensional Central Limit Theorem for the sum of random vectors in the plane (orthogonal case) or in three-dimensional space (unitary case). In particular, *only for the orthogonal case*, the limiting situations $N = 2, M \gg 1$ and $M = 2, N \gg 1$ yield identical spacing distributions $p(s)$ (after a suitable rescaling), of the standard Wigner’s surmise form (A.2) for $\beta = 1$. For the unitary case, it would be interesting to understand in more detail how the transition from the scaling function $\Phi_{\mathbf{U}}(x)$ (107) to the Wigner’s surmise (A.2) for $\beta = 2$ occurs upon increasing the size N . Distributions of eigenvalues with infinite variance, for which the Central Limit Theorem breaks down, may lead to different scaling functions and this constitutes another possible direction for further research.

Our simple construction offers analytical access to the microscopic statistics of the free addition of random matrices for both small and large N , a task that is usually considered to be very hard to accomplish with standard tools of free probability. All the results have been corroborated by numerical simulations. Our work can stimulate fur-

ther research in the following directions: on one hand, it would be interesting to analyze in more detail the jpd (4) of our model and its marginals (average density and correlation functions). On the other hand, further analytical insight into the limit $N \rightarrow \infty$ could be perhaps gained by analyzing the large- N limit of the HCIZ integrals appearing in 5 [16]. It will also be interesting to see if more exotic choices for the distributions $p^{(1,2)}(x)$ (for example yielding a $\omega^{(1)}(0) \rightarrow \infty$) may further change the spacing distribution for $s \rightarrow 0^+$. The limit $M \gg 1$ at fixed (small) $N > 2$ is also interesting and is left for future investigation. Finally, the recently introduced *spacing ratio distribution* [19, 20] will be well worth studying in our model, for example for $N = 3$ (the smallest sensible size). This should then be compared with the classical GOE and GUE results, which are known to be highly universal, in particular regarding the role played by the number of summands M .

Acknowledgments

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Appendix A. Level spacings and correlation functions

Given a sequence of eigenvalues $\{\lambda_i\}_{i=1,\dots,N}$ of a given random matrix, the statistics of distances between neighboring eigenvalues, the so called eigenvalue spacing distribution, is the benchmark observable when studying the short-range statistical properties of a given random matrix ensemble’s average spectrum. Assuming that the eigenvalues have been sorted ($\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$), one would be tempted to study the statistics of the distances $\lambda_{i+1} - \lambda_i$. However, such a naive implementation would completely overlook some of the universal features shared by several random matrix ensembles. These are “hidden” by the non-constant density of the eigenvalues, which pushes them too close or pulls them too far apart in different regions of the spectrum. Indeed, universal properties are recovered once the eigenvalue density has been converted into a constant one. This can be achieved by means of a procedure known in the literature as *unfolding*. In a nutshell, the unfolded levels for sufficiently large matrices are defined as $\xi_i = NF(\lambda_i)$, where

$$F(x) = \int_{-\infty}^x d\lambda \rho_{\infty}(\lambda) \quad (\text{A.1})$$

is the cumulative distribution corresponding to the eigenvalue density $\rho_{\infty}(\lambda)$. Of course, such a procedure is rather easy to perform whenever the density $\rho_{\infty}(\lambda)$ is known in closed form. However, this is not the case in most practical situations, and one typically has to resort to numerical methods to empirically estimate the unknown spectral density, for example by using a kernel smoothing method [21]. Once the unfolded eigenvalues ξ_i

have been obtained, one eventually gets the distribution of the spacings $s_i = \xi_{i+1} - \xi_i$, which by definition has mean $\langle s \rangle = 1$. For general N , the spacing distribution has a complicated expression [1], which is however fairly well approximated by an exact calculation for $N = 2$, the so called Wigner's surmise

$$p_{WS}^{(\beta)}(s) = a_\beta s^\beta e^{-b_\beta s^2}, \quad (\text{A.2})$$

where $a_\beta = 2 \Gamma^{\beta+1}((\beta+2)/2)/\Gamma^{\beta+2}((\beta+1)/2)$, $b_\beta = \Gamma^2((\beta+2)/2)/\Gamma^2((\beta+1)/2)$, and $\beta = 1, 2$ for real symmetric and complex hermitian matrices, respectively. These are the curves we plotted in Fig. 2 and 3 for comparison.

Unfolded eigenvalues can be used also to empirically estimate correlation functions of random matrix ensembles, which in turn probe the long-range properties of eigenvalue spectra. Given a random matrix ensemble described by an eigenvalue joint probability density $\mathcal{P}(\lambda_1, \dots, \lambda_N)$, *e.g.* the one in equation (1), its generic n -point correlation function is defined as

$$R_n(\lambda_1, \dots, \lambda_n) = \frac{N!}{(N-n)!} \int d\lambda_{n+1} \dots d\lambda_N \mathcal{P}(\lambda_1, \dots, \lambda_N). \quad (\text{A.3})$$

In this paper we only considered one-point and two-point correlation functions (see Fig. 1): the former essentially coincides with the average eigenvalue density ($R_1(\lambda) = N\rho_N(\lambda)$, see equation (2)), whereas the latter is typically used to compute the expected number of eigenvalue pairs in a given interval $\mathcal{I}$ as

$$\int_{\mathcal{I} \times \mathcal{I}} d\lambda_1 d\lambda_2 R_2(\lambda_1, \lambda_2). \quad (\text{A.4})$$

Now, the two-point correlation function for the GUE (or any invariant ensemble with $\beta = 2$) can be written as

$$R_2(\lambda_1, \lambda_2) = K_N(\lambda_1, \lambda_1)K_N(\lambda_2, \lambda_2) - [K_N(\lambda_1, \lambda_2)]^2, \quad (\text{A.5})$$

where K_N is the ensemble's *kernel* (see *e.g.* [1, 2]). Now, in the large N limit one can verify [22] that, after a few manipulations, the above equation can be written as

$$R_2(\lambda_1, \lambda_2) = 1 - [K_\infty(N\rho_\infty(\lambda_2)(\lambda_2 - \lambda_1))]^2, \quad (\text{A.6})$$

where the density $\rho_\infty(\lambda)$ was introduced in equation (2), and (rather universally) $K_\infty(r) = \sin(\pi r)/(\pi r)$. Assuming that $\lambda_2 = \lambda_1 + h$ (with $h > 0$), the kernel's argument in the above equation becomes

$$\begin{aligned} N\rho(\lambda_2)(\lambda_2 - \lambda_1) &= N\rho_\infty(\lambda_2)h = NF'(\lambda_2)h \sim N(F(\lambda_2) - F(\lambda_2 - h)) \\ &= N(F(\lambda_2) - F(\lambda_1)) = \xi_2 - \xi_1, \end{aligned} \quad (\text{A.7})$$

where $F(\lambda)$ is defined in (A.1), whereas ξ_i denotes the i -th unfolded eigenvalue. Thus, the two-point correlation function effectively describes the statistics of distances between unfolded eigenvalues. However, equation (A.6) only holds where the kernel K_∞ accurately describes the eigenvalues' properties, *i.e.* in the bulk of the spectrum where $h = \lambda_2 - \lambda_1 \sim \mathcal{O}(1/N)$ ($\xi_2 - \xi_1 \sim 1$ on the unfolded scale). In such a limit, the kernel

does not depend on the specific positions of the eigenvalues, so that for the GUE we have:

$$R_2(r) = 1 - \left(\frac{\sin(\pi r)}{\pi r} \right)^2, \quad r = \xi_2 - \xi_1. \quad (\text{A.8})$$

Equation (A.7) and the following considerations mean that the correlation function (A.8) has to be compared with sampled (from large matrices) distances of order unity on the unfolded eigenvalue scale, and this is why the numerical check in the bottom panel of Fig. 1 is plotted only up to $r = 2.5$ ¶.

The mathematical details of the derivation of the two-point correlation function for the GOE are rather different from the ones sketched above and much more involved, so we shall not report them here (see [1, 2] for further details). The final result reads:

$$R_2(x) = 1 - \left(\frac{\sin(\pi x)}{\pi x} \right)^2 - \left(\frac{\cos(\pi x)}{x} - \frac{\sin(\pi x)}{\pi x^2} \right) \left(\frac{1}{2} - \frac{\text{Si}(\pi x)}{\pi} \right), \quad (\text{A.9})$$

where $\text{Si}(x) = \int_0^x dt \sin(t)/t$ is the sine integral, and it can be numerically compared to distances of unfolded eigenvalues following the same criteria mentioned above.

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¶ From an operational point of view, a numerical check such as the one in Fig. 1 can be performed by computing the distances between unfolded eigenvalues of matrices of size N , and retaining only those that are smaller than a certain threshold of order one (2.5 in the example of Fig. 1). If this exercise is repeated for L independent matrices, then the correlation function (A.8) can be matched by initially producing the (non-normalized) histogram of all the selected (unfolded) distances, and subsequently dividing it by $NL\Delta$, where Δ is the bin width. The division by NL ensures that the bins of the histogram represent a proxy for the probability of finding a pair of eigenvalues separated by a distance range specified by the particular bin (conditional on such a distance being smaller than the chosen threshold), whereas the division by Δ eventually adjusts the area under the histogram to the area underlying the correlation function curve on the selected support.

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